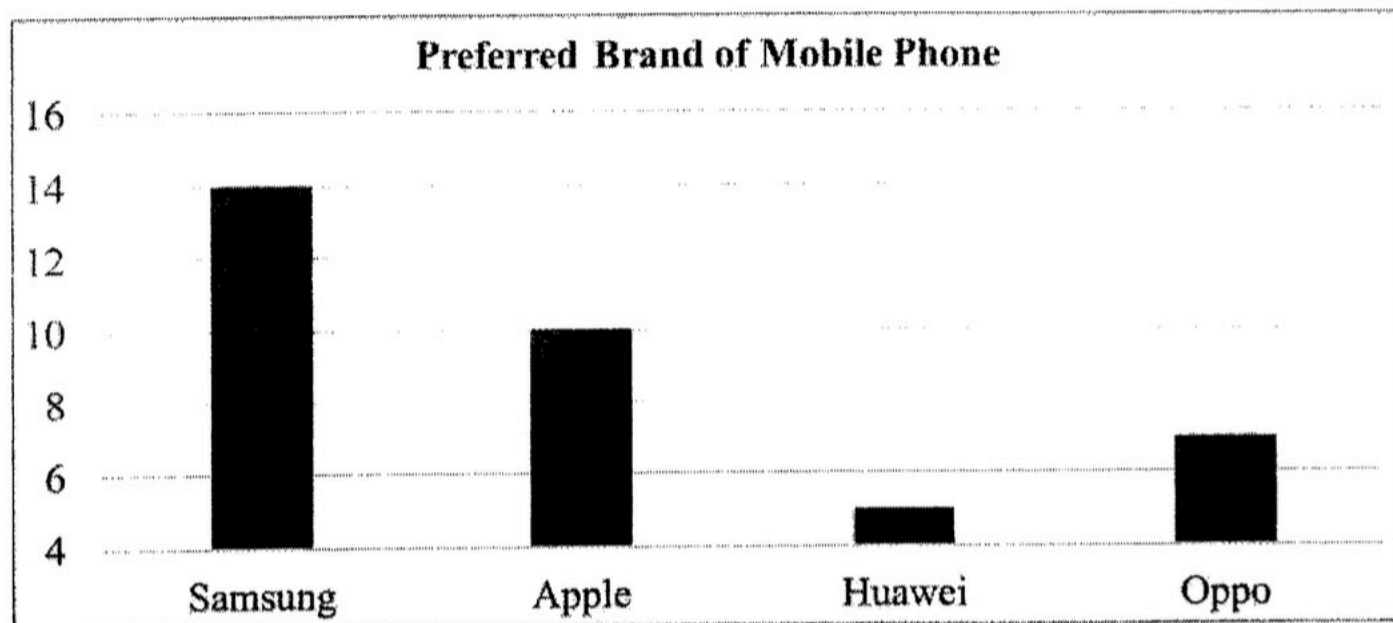


- 4 A survey was conducted to find out the preferred brand of mobile phone that secondary school students like.



State one aspect of the graph that may be misleading and explain how this may lead to a misinterpretation of the graph.

.....

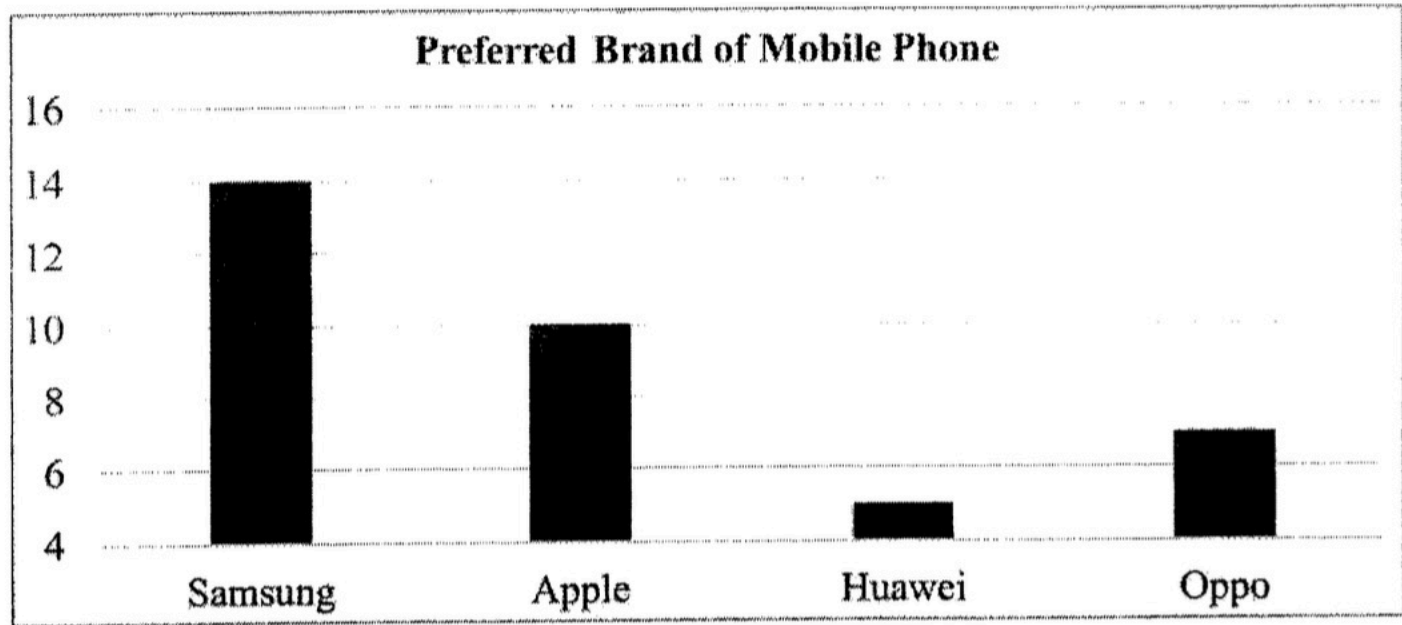
.....

.....[2]

-
- 5 Solve the inequality $-2+x \leq 2-3x < 9+4x$.

Answer [2]

- 4 A survey was conducted to find out the preferred brand of mobile phone that secondary school students like.



State one aspect of the graph that may be ^{1m}misleading and ^{1m}explain how this may lead to a misinterpretation of the graph.

state why is misleading

The y-axis does not start from 0. [1m]

explain why is misleading

Hence it looks like there are 3 times the no. of students

choosing Samsung than Oppo, which is in fact [2]

only 2 times. [1m]

- 5 Solve the inequality $-2+x \leq 2-3x < 9+4x$.

$$-2+x \leq 2-3x$$

$$-4 \leq -4x$$

$$4 \geq 4x$$

$$1 \geq x$$

$$x \leq 1$$



$$\therefore -1 < x \leq 1$$

$$2-3x < 9+4x$$

$$-3x-4x < 9-2$$

$$-7x < 7$$

$$x > -1$$

Answer $-1 < x \leq 1$ [2]

7 Factorise $4ax^2 - 4ay^2 + 2bx^2 - 2by^2$ completely.

Answer[2]

7 Factorise $4ax^2 - 4ay^2 + 2bx^2 - 2by^2$ completely.

$$\begin{aligned} & 4ax^2 - 4ay^2 + 2bx^2 - 2by^2 \\ &= 4a(x^2 - y^2) + 2b(x^2 - y^2) \\ &= (4a + 2b)(x^2 - y^2) \quad \Rightarrow a^2 - b^2 = (a+b)(a-b) \quad \mathbf{B1: (x+y)(x-y)} \\ &= 2(2a + b)(x - y)(x + y) \quad \mathbf{B1} \end{aligned}$$

Answer[2]

8 Simplify $\left(\frac{a^{16}}{b^{-6}}\right)^{\frac{3}{2}}$.

Answer[2]

9 Given that $25 \times 125^x = 5$, find the value of x .

Answer $x =$ [2]

8 Simplify $\left(\frac{a^{16}}{b^{-6}}\right)^{\frac{3}{2}}$.

$= (a^{16} b^6)^{\frac{3}{2}}$

$= a^{16(\frac{3}{2})} b^{6(\frac{3}{2})} \quad \Rightarrow (a^m)^n = a^{mn}$

$= a^{24} b^9$

Answer $a^{24} b^9$ [2]

9 Given that $25 \times 125^x = 5$, find the value of x .

$5^2 \times (5^3)^x = 5$

$5^2 \times 5^{3x} = 5$

$5^{2+3x} = 5$

$\Rightarrow a^m \times a^n = a^{m+n}$

comparing index,

$2 + 3x = 1$

$3x = -1$

$x = -\frac{1}{3}$

Answer $x =$ $-\frac{1}{3}$ [2]

10 Solve $\frac{5}{(2-x)^2} + \frac{1}{x-2} = 4$.

Answer $x = \dots\dots\dots$ [3]

11 (a) Express $1+x^2-5x$ in the form of $(x+a)^2+b$.

Answer $\dots\dots\dots$ [1]

(b) Write down the equation of line of symmetry of the graph of $y=1+x^2-5x$.

Answer $\dots\dots\dots$ [1]

(c) Write down the coordinates of intersection between the line of symmetry and the graph of $y=1+x^2-5x$.

Answer $(\dots\dots\dots, \dots\dots\dots)$ [1]

10 Solve $\frac{5}{(2-x)^2} + \frac{1}{x-2} = 4$.

$$\frac{5}{(2-x)^2} + \frac{1}{x-2} = 4$$

$$\frac{5}{(x-2)^2} + \frac{1}{x-2} = 4$$

$$\frac{5}{(x-2)^2} + \frac{x-2}{(x-2)^2} = 4$$

$$\frac{5+x-2}{(x-2)^2} = 4$$

$$\frac{3+x}{(x-2)^2} = 4$$

$$3+x = 4(x-2)^2$$

$$3+x = 4(x^2 - 4x + 4)$$

$$3+x = 4x^2 - 16x + 16$$

$$4x^2 - 17x + 13 = 0$$

$$(x-1)(4x-13) = 0$$

$$x = 1 \text{ or } x = 3.25$$

M1: Common denominator (only accept quadratic denominator)

M1: Factorisation or quadratic formula

A1

Answer $x = \dots\dots\dots$ [3]

11 (a) Express $1+x^2-5x$ in the form of $(x+a)^2+b$.

$$1+x^2-5x = x^2-5x+1$$

$$= \left(x - \frac{5}{2}\right)^2 - \left(\frac{5}{2}\right)^2 + 1$$

$$= (x-2.5)^2 - 5.25$$

$$= [x+(-2.5)]^2 + (-5.25)$$

B1: Must be this form

Answer $\dots\dots\dots$ [1]

(b) Write down the equation of line of symmetry of the graph of $y = 1+x^2-5x$.

Eqn of line of symmetry: $x = 2.5$

B1

$$y = x^2 - 5x + 1$$

Quadratic graph

Answer $\dots\dots\dots$ [1]

Relap:

$$y = x^2$$

positive

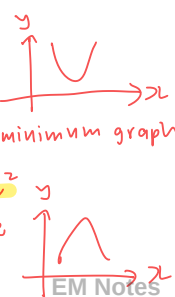
while $\Rightarrow$ minimum graph

$$y = -x^2$$

negative

from

maximum graph



u

- (c) Write down the coordinates of intersection between the line of symmetry and the graph of $y = 1 + x^2 - 5x$. $y = (x - 2.5)^2 - 5.25$

Point of intersection is at the turning point.

$(2.5, -5.25)$

 **B1**

To find max/min point, make $(x - 2.5) = 0$
hence, sub $x = 2.5$.

$$y = (0)^2 - 5.25 = -5.25$$

Answer (.....,) [1]

14 A map of the United States of America has a scale of 1 : 8 000 000 .

- (a) The length of the Mississippi River is 3766 km.
Calculate the length, in centimetres, of Mississippi River on the map.

Answercm [1]

- (b) The area of California on the map is 66.25 cm².
Calculate the actual area, in square kilometres, of California.

Answerkm² [2]

14 A map of the United States of America has a scale of 1 : 8 000 000 .

(a) The length of the Mississippi River is 3766 km.

Calculate the length, in centimetres, of Mississippi River on the map.

$$1 \text{ cm} : 8000000 \text{ cm}$$

$$1 \text{ cm} : 80 \text{ km}$$

$$80 \text{ km} \rightarrow 1 \text{ cm}$$

$$3766 \text{ km} \rightarrow \frac{3766}{80} = 47.075 \text{ cm}$$

$$\mathbf{B1: 47.075}$$

Answercm [1]

(b) The area of California on the map is 66.25 cm².

Calculate the actual area, in square kilometres, of California.

$$1 \text{ cm} : 8000000 \text{ cm}$$

M1: Using proportion

$$(1 \text{ cm})^2 : (80 \text{ km})^2$$

$$\therefore 1 \text{ cm}^2 : 6400 \text{ km}^2$$

$$1 \text{ cm}^2 \rightarrow 6400 \text{ km}^2$$

$$66.25 \text{ cm}^2 \rightarrow 66.25(6400) = 424000 \text{ km}^2$$

$$\mathbf{A1}$$

Answerkm² [2]

17 $\xi = \{\text{integers } x : 1 < x < 21\}$

$A = \{\text{integers that are perfect squares}\}$

$B = \{\text{integers that are not prime numbers}\}$

$C = \{\text{integers that are divisible by 5}\}$

(a) List down all the elements in A , B and C .

Answer $A = \dots\dots\dots$

$B = \dots\dots\dots$

$C = \dots\dots\dots[2]$

A number is chosen randomly from ξ .

(b) Find the number of elements in $B \cap C'$.

Answer $\dots\dots\dots [1]$

(c) Hence, find the probability that the chosen number is not a prime number and is not divisible by 5.

Answer $\dots\dots\dots [1]$

17 $\xi = \{\text{integers } x : 1 < x < 21\}$

$A = \{\text{integers that are perfect squares}\}$

$B = \{\text{integers that are not prime numbers}\}$

$C = \{\text{integers that are divisible by 5}\}$

(a) List down all the elements in A , B and C .

$A = \{4, 9, 16\}$

$B = \{4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20\}$

$C = \{5, 10, 15, 20\}$

B2

Minus one mark per mistake

Answer $A = \dots\dots\dots$

$B = \dots\dots\dots$

$C = \dots\dots\dots$ [2]

A number is chosen randomly from ξ .

(b) Find the number of elements in $B \cap C'$.

$B = \{4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20\}$

$C = \{5, 10, 15, 20\}$

$B \cap C' = \{4, 6, 8, 9, 12, 14, 16, 18\}$

$\therefore$ Required number = 8

B1

Answer $\dots\dots\dots$ [1]

(c) Hence, find the probability that the chosen number is not a prime number and is not divisible by 5.

Required probability = $\frac{8}{19}$

B1: Follow through from part (b) answer, but denominator must be 19

Answer $\dots\dots\dots$ [1]

- 18 (a) The prime factorisation of 450 is $2 \times 3^2 \times 5^2$.
Express 648 as a product of its prime factors.

Answer $648 = \dots\dots\dots$ [1]

- (b) Using your answer to **part (a)**, determine whether 583200 is a perfect square.

$\dots\dots\dots$
 $\dots\dots\dots$
 $\dots\dots\dots$ [2]

- (c) Find the smallest integer n such that $450n$ is a multiple of 648.

Answer $n = \dots\dots\dots$ [1]

18 (a) The prime factorisation of 450 is $2 \times 3^2 \times 5^2$.

Express 648 as a product of its prime factors.

$$648 = 2^3 \times 3^4$$

B1

Answer 648 =[1]

(b) Using your answer to part (a), determine whether 583200 is a perfect square.

Note that $450 \times 648 = 291600$

$$\therefore 583200 = 2 \times 450 \times 648 = 2^5 \times 3^6 \times 5^2$$

B1

583200 is not a perfect square as the power (exponent) of base 2 in 583200 is not an even number.

B1

.....

.....

.....[2]

(c) Find the smallest integer n such that $450n$ is a multiple of 648.

$$n = \frac{LCM(450, 648)}{450}$$

$$= \frac{2^3 \times 3^4 \times 5^2}{2 \times 3^2 \times 5^2}$$

$$= 2^2 \times 3^2$$

$$= 36$$

B1

Answer $n =$ [1]

20 It is given that $\overrightarrow{AB} = \begin{pmatrix} -12 \\ a \end{pmatrix}$, $a > 0$ and $|\overrightarrow{BA}| = 37$ units.

(a) Find the value of a .

Answer $a = \dots\dots\dots$ [2]

(b) C is a point $(0,10)$ and $\overrightarrow{AB} = \overrightarrow{DC}$.

Find the coordinates of D .

Answer $(\dots\dots\dots, \dots\dots\dots)$ [2]

(c) What type of quadrilateral is $ABCD$? Explain your answer.

.....

[1]

20 It is given that $AB = \begin{pmatrix} -12 \\ a \end{pmatrix}$, $a > 0$ and $|BA| = 37$ units. use magnitude formula.
 $|\vec{AB}| = \sqrt{x^2 + y^2}$

(a) Find the value of a .

$$|BA| = |AB| = 37$$

$$\left| \begin{pmatrix} -12 \\ a \end{pmatrix} \right| = 37$$

$$\sqrt{(-12)^2 + a^2} = 37$$

$$144 + a^2 = 1369$$

$$a^2 = 1225$$

$$a = 35 \text{ or } a = -35 \text{ (NA)}$$

M1: Distance formula

A1

Answer $a = \dots\dots\dots[2]$

(b) C is a point $(0,10)$ and $\overline{AB} = \overline{DC}$.

Find the coordinates of D .

$$\overline{AB} = \overline{DC}$$

$$\begin{pmatrix} -12 \\ 35 \end{pmatrix} = \overline{OC} - \overline{OD}$$

M1: $OC - OD$

$$\overline{OD} = \overline{OC} - \begin{pmatrix} -12 \\ 35 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \\ 10 \end{pmatrix} - \begin{pmatrix} -12 \\ 35 \end{pmatrix}$$

$$= \begin{pmatrix} 12 \\ -25 \end{pmatrix}$$

$\therefore$ Coordinates of $D = (12, -25)$

A1

Answer (.....,)[2]

(c) What type of quadrilateral is $ABCD$? Explain your answer.

$$AB = DC$$

There is a pair of parallel sides with equal length.

$\therefore ABCD$ is a parallelogram.

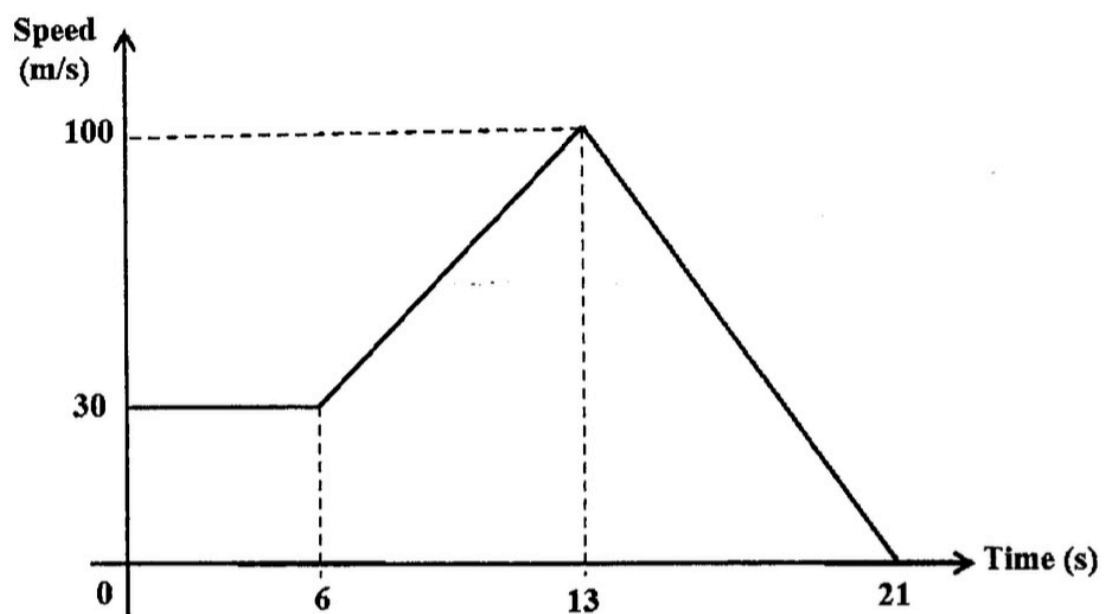
B1: Parallelogram

.....

.....

.....[1]

- 22 The diagram shows the speed-time graph of a particle.



- (a) Find the speed of the particle when $t = 12\text{s}$.

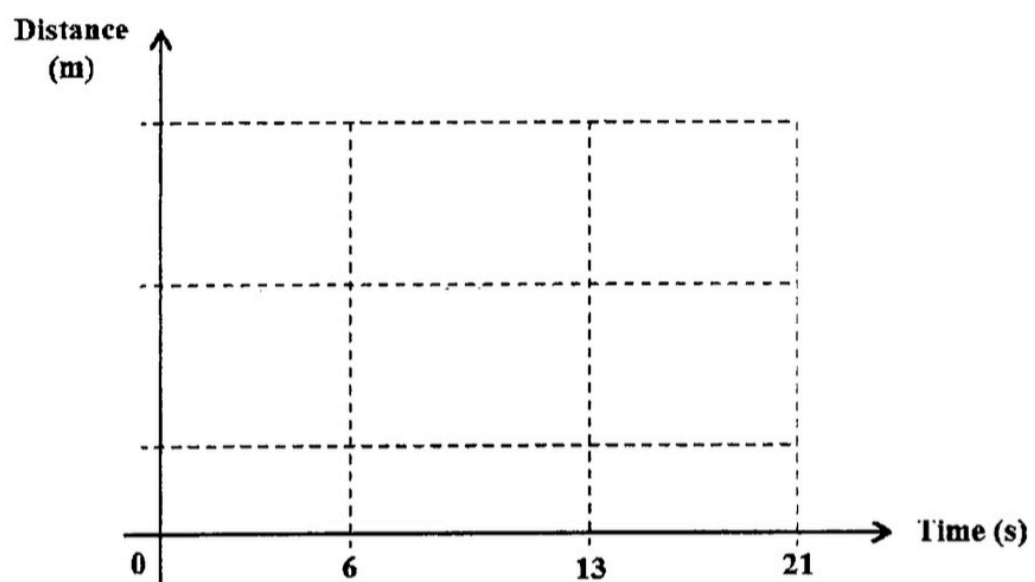
Answerm/s [2]

- (b) Find the distance covered before the particle starts to slow down.

Answerm [2]

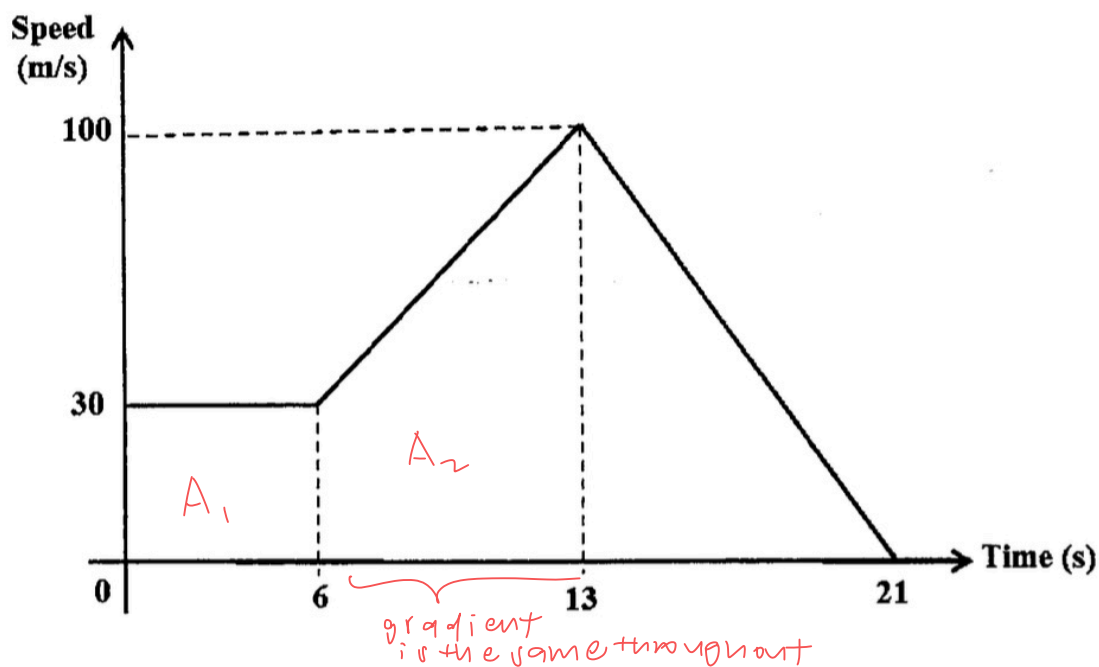
(c) Complete the distance-time graph of the particle.

[1]



→ revise speed-time/distance-time graph (under unit 1.4 of revision booklet)

22 The diagram shows the speed-time graph of a particle.



(a) Find the speed of the particle when $t = 12$ s.

Let unknown speed be v .

$$\frac{v-30}{12-6} = \frac{100-30}{13-6} \Rightarrow \text{use grad. formula } \frac{y_1-y_2}{x_1-x_2}$$

$$\frac{v-30}{6} = 10$$

$$v-30 = 60$$

$$v = 90 \text{ m/s}$$

Answer 90 m/s [2]

(b) Find the distance covered before the particle starts to slow down.

→ In a speed-time graph, distance = Area under the graph

$$A_1 = 30 \times 6 \\ = 180 \text{ m}$$

$$A_2 = \frac{1}{2} (30 + 100) (13 - 6) \\ = 455 \text{ m}$$

$$\text{Total distance} = 180 + 455 \\ = 635 \text{ m}$$

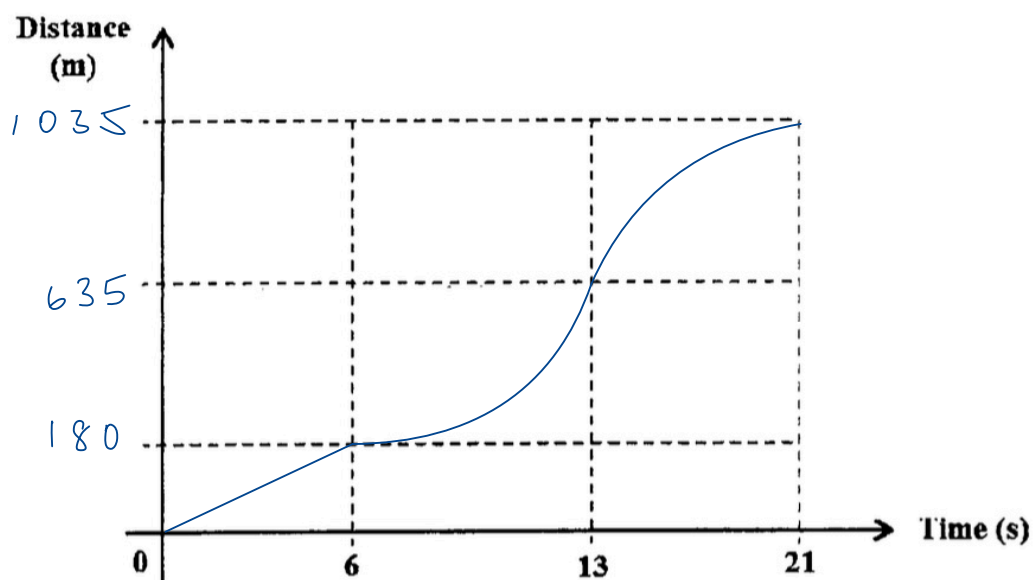
Answer 635 m [2]

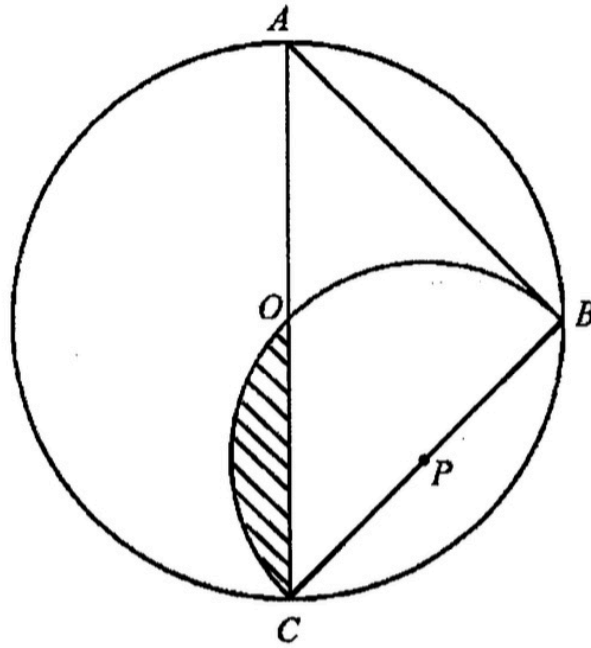


★ trapezium formula $\frac{1}{2} \times (a+b) \times h$

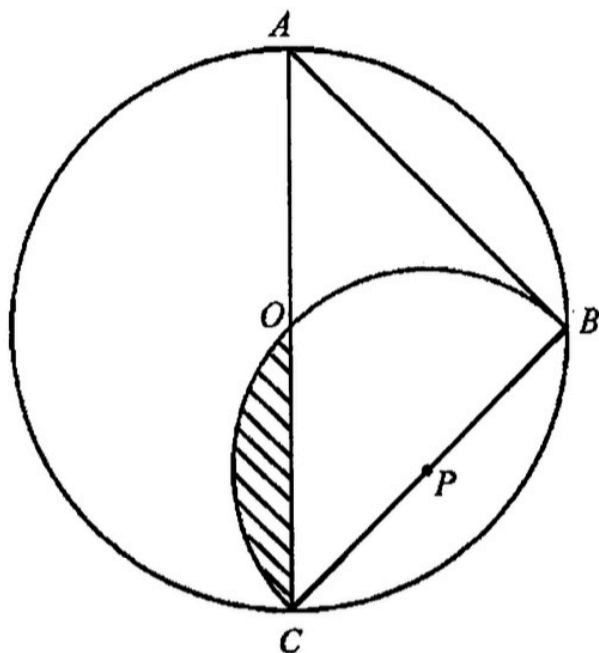
(c) Complete the distance-time graph of the particle.

[1]





AC is a diameter of a circle with centre O . $AC = \sqrt{2}r$ and $AB = BC$.
 COB is a semi-circular arc, centre P .
 What percentage of the circle is **not** shaded?



AC is a diameter of a circle with centre O . $AC = \sqrt{2}r$ and $AB = BC$.

COB is a semi-circular arc, centre P .

What percentage of the circle is **not** shaded?

$\angle ABC = 90^\circ$ (Right angle in semicircle) **B1:** Need to show $\angle ABC = 90^\circ$

$$AB = BC,$$

$$(AB)^2 + (BC)^2 = (\sqrt{2}r)^2$$

$$2(BC)^2 = 2r^2$$

$$(BC)^2 = r^2$$

M1: Finding AB or BC

$$\therefore BC = r \text{ (Reject -ve)}$$

$$OC = \frac{1}{2} AC = \frac{\sqrt{2}}{2} r$$

$$CP = PO = \frac{1}{2} BC = \frac{r}{2}$$

$$OC^2 = CP^2 + PO^2$$

$\therefore$ By converse of Pythagoras' Theorem,

Method 1: $\triangle OPC$ is a right angled $\triangle$.

B1: Need to show

Method 2:

$\triangle OPC$ is a right angled $\triangle$

Note that P and O are midpoints of BC and AC respectively.

$\therefore$ By midpoint theorem,

OP is parallel to AB ,

and $\angle OPC = \angle ABC$

$\angle ABC = 90^\circ$ (Right angle in semicircle)

$\therefore \angle OPC = 90^\circ$

$\therefore \triangle OPC$ is a right angled Δ .

M1: Using midpoint theorem

A1: $\angle OPC = \angle ABC$

B1: Need to show $\triangle OPC$ is a right angled Δ

Percentage of circle shaded

$$= \frac{\text{Sector OPC} - \text{Triangle OPC}}{\text{Area of big circle}} \times 100\%$$

$$= \frac{\frac{1}{2} \left(\frac{r}{2} \right)^2 \left(\frac{\pi}{2} \right) - \frac{1}{2} \left(\frac{r}{2} \right)^2 \sin \left(\frac{\pi}{2} \right)}{\pi \left(\frac{\sqrt{2}}{2} r \right)^2} \times 100\%$$

$$= \frac{\frac{1}{16} \pi r^2 - \frac{1}{8} r^2}{\frac{1}{2} \pi r^2} \times 100\%$$

$$= \frac{\frac{1}{16} \pi - \frac{1}{8}}{\frac{1}{2} \pi} \times 100\%$$

$$= 4.542252845\%$$

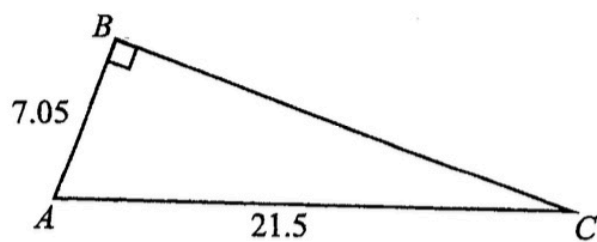
$$\begin{aligned} \text{Percentage of circle shaded} &= 100\% - 4.542252845\% \\ &= 95.5\% \text{ (3 s.f)} \end{aligned}$$

M1: Correct percentage formula with the relevant shapes found using the correct formula (FT)

A1

Answer% [5]

3

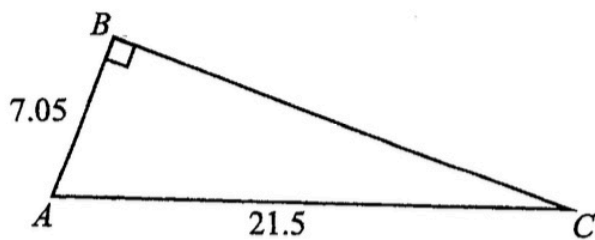


In the triangle, $AC = 21.5$ cm, $AB = 7.05$ cm and angle $ABC = 90^\circ$.

Calculate BC .

Answer $BC = \dots\dots\dots$ cm [2]

3



In the triangle, $AC = 21.5$ cm, $AB = 7.05$ cm and angle $ABC = 90^\circ$.

Calculate BC .

$$BC^2 = 21.5^2 - 7.05^2 \text{ (Pythagoras' Theorem)}$$

$$BC = \sqrt{412.5475}$$
$$= 20.3 \text{ cm (3.s.f.)}$$

Answer $BC = 20.3$ cm [2]

4

- 4 Shaqil invests \$3500 at a simple interest of 2.3% per year for 5 years.

Calculate the total value of his investments at the end of 5 years.

Answer \$ [2]

- 5 p is inversely proportional to $\sqrt[3]{q}$.
The value of q is decreased by 87.5%.

Calculate the percentage increase in the value of p .

Answer [2]

$$I = \frac{PRT}{100}$$

- 4 Shaqil invests \$3500 at a simple interest of 2.3% per year for 5 years.

Calculate the total value of his investments at the end of 5 years.

$$I = \frac{3500(2.3)(5)}{100}$$

$$= \$402.50$$

$$\text{Total investment} = \$3500 + \$402.50$$

$$= \$3902.50$$

Answer \$ 3902.50 [2]

- 5 p is inversely proportional to $\sqrt[3]{q}$.
The value of q is decreased by 87.5%.

Calculate the percentage increase in the value of p .

$$P_o = \frac{k}{\sqrt[3]{q}}$$

$$P_N = \frac{k}{\sqrt[3]{0.125q}}$$

$$P_N = \frac{k}{0.5\sqrt[3]{q}}$$

$$P_N = \frac{2k}{\sqrt[3]{q}}$$

compare b/w
old p value &
new p value

$$\% \text{ increase} = \frac{\text{new value} - \text{old value}}{\text{old value}} \times 100\%$$

$$= \frac{2P_o - P_o}{P_o} \times 100\%$$

$$= \frac{P_o}{P_o} \times 100\%$$

$$= 1 \times 100\%$$

$$= 100\%$$

$$P_N = 2P_o$$

Answer 100% [2]

6

- 8 The volume of a cuboid is 1296 cm^3 .
The area of the **largest** face is 162 cm^2 .
The dimensions of the cuboid have integer values.

Find the dimensions of the cuboid.

Answercm bycm by.....cm [2]

6

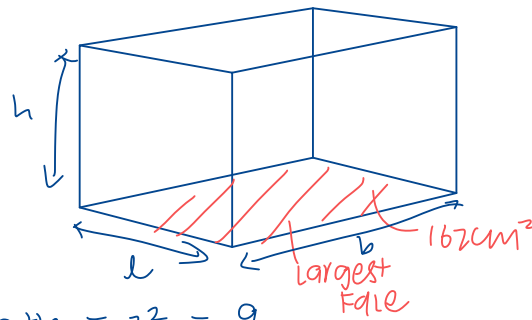
- 8 The volume of a cuboid is 1296 cm^3 .
The **area of the largest face** is 162 cm^2 .
The dimensions of the cuboid have integer values.

Find the dimensions of the cuboid.

$$1296 = 2^4 \times 3^4$$

$$162 = 2 \times 3^4$$

$$\begin{aligned} \text{volume} &= BA \times h \\ 1296 &= 162 \times h \\ h &= \frac{1296}{162} \\ &= 8 \end{aligned}$$



$$\begin{aligned} \text{length} &= 3^2 = 9 \\ \text{breadth} &= 3^2 \times 2 = 18 \end{aligned}$$

Answer 9 cm by 18 cm by 8 cm [2]

7

10 (a) Solve $\frac{5}{x} + 14 = 7$.

Answer [1]

(b) Express as a single fraction in its simplest form $\frac{3x-2}{5} - \frac{x+1}{4}$.

Answer [2]

10 (a) Solve $\frac{5}{x} + 14 = 7$.

$$\frac{5}{x} = -7$$

$$5 = -7x$$

$$x = -\frac{5}{7}$$

Answer $x = -\frac{5}{7}$ [1]

(b) Express as a single fraction in its simplest form $\frac{3x-2}{5} - \frac{x+1}{4}$.

$$= \frac{4(3x-2)}{20} - \frac{5(x+1)}{20}$$

$$= \frac{12x-8-5x-5}{20}$$

$$= \frac{7x-13}{20}$$

Answer $\frac{7x-13}{20}$ [2]

9

13 In a regular polygon, the ratio interior angle : exterior angle = 7 : 2.

Calculate the number of sides of the polygon.

Answer [3]

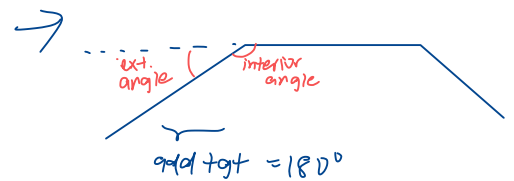
9

13 In a regular polygon, the ratio interior angle : exterior angle = 7 : 2.

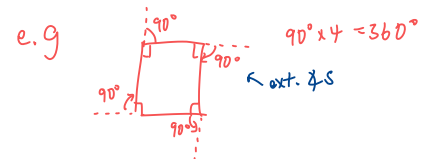
Calculate the number of sides of the polygon.

$$\begin{aligned} \text{exterior angle} &= (180 \div 9) \times 2 \\ &= 40^\circ \end{aligned}$$

$$\begin{aligned} \text{number of sides of polygon} &= \frac{360^\circ}{40^\circ} \\ &= 9 \end{aligned}$$

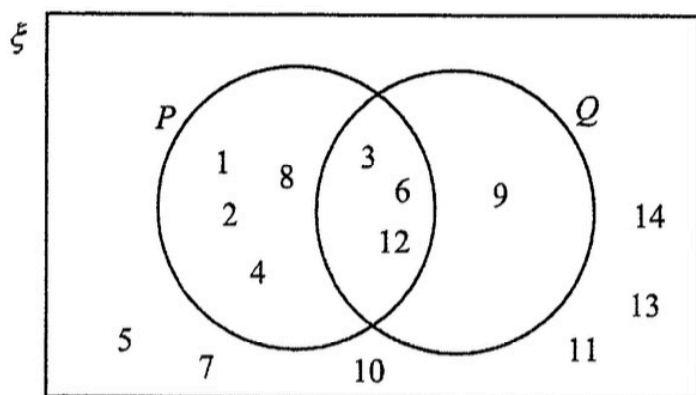


★ must know.
FOR ALL POLYGON,
the exterior ∠s add up to 360°



Answer 9 [3]

16 The Venn diagram shows the elements of $\xi = \{\text{integers } x : 1 \leq x \leq 14\}$.



(a) Underline the correct statement from the list below.

$$(P \cup Q)' = \emptyset$$

$$P \cup Q = \{3, 6, 9\}$$

$$P' \subset Q$$

$$10 \notin P \cap Q$$

$$P' \cap Q = \{5, 7, 10, 11, 13, 14\}$$

[1]

(b) Find the number of elements in

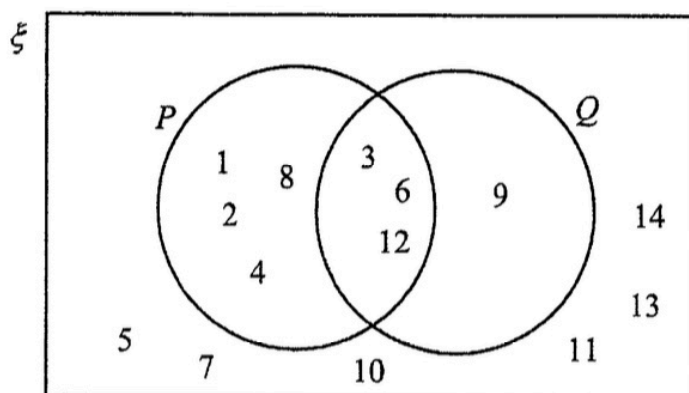
(i) $(P \cup Q)'$,

Answer [1]

(ii) $(P \cup Q) \cap (P \cap Q)'$.

Answer [1]

16 The Venn diagram shows the elements of $\xi = \{\text{integers } x: 1 \leq x \leq 14\}$.



(a) Underline the correct statement from the list below.

$$(P \cup Q)' = \emptyset$$

$$P \cup Q = \{3, 6, 9\}$$

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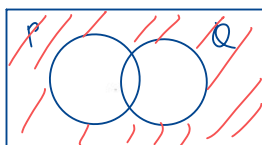
$$P' \cap Q = \{5, 7, 10, 11, 13, 14\}$$

[1]

(b) Find the **number of elements in**

(i) $(P \cup Q)'$,

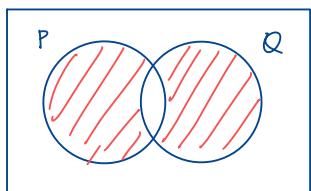
$$n(P \cup Q)' = 6$$



Answer 6 [1]

(ii) $(P \cup Q) \cap (P \cap Q)'$.

$$n(P \cup Q) \cap (P \cap Q)' = 5$$



Answer 5 [1]

19 (a) Express $x^2 - 11x + 13$ in the form of $(x + a)^2 + b$.

Answer [1]

(b) Hence, solve the equation $x^2 - 11x + 13 = 0$, giving your answers to 2 decimal places.

Answer [3]

19 (a) Express $x^2 - 11x + 13$ in the form of $(x+a)^2 + b$.

$$= \left(x - \frac{11}{2}\right)^2 - \left(-\frac{11}{2}\right)^2 + 13$$

$$= \left(x - \frac{11}{2}\right)^2 - \frac{69}{4}$$

Answer $\left(x - \frac{11}{2}\right)^2 - \frac{69}{4}$ [1]

must use above answer

(b) Hence, solve the equation $x^2 - 11x + 13 = 0$, giving your answers to 2 decimal places.

$$\left(x - \frac{11}{2}\right)^2 - \frac{69}{4} = 0$$

$$\left(x - \frac{11}{2}\right)^2 = \frac{69}{4}$$

$$x - \frac{11}{2} = \pm \sqrt{\frac{69}{4}}$$

must $\pm \sqrt{\quad}$

$$x = \sqrt{\frac{69}{4}} + \frac{11}{2} \text{ or } x = -\sqrt{\frac{69}{4}} + \frac{11}{2}$$

$$= 9.65 \text{ (2.d.p)} = 1.35 \text{ (2.d.p)}$$

Answer $x = 9.65 \text{ or } x = 1.35$ [3] *2M*

- 21 (a) Two integers x and y can be expressed as a product of their prime factors as shown below.

$$x = 2^3 \times 3^n \times 5$$

$$y = 2^2 \times 3^2 \times 5^m$$

The lowest common multiple (LCM) of x and y is 360.

The highest common factor (HCF) of x and y is 60.

Find the values of n and m .

Answer $n = \dots\dots\dots$

$m = \dots\dots\dots$ [2]

- (b) The square root of an integer z is equal to the cube of $3^2 \times 5^4$.

Write z as a product of its prime factors.

Answer $\dots\dots\dots$ [2]

- 21 (a) Two integers x and y can be expressed as a product of their prime factors as shown below.

$$x = 2^3 \times 3^n \times 5$$

$$y = 2^2 \times 3^2 \times 5^m$$

The lowest common multiple (LCM) of x and y is 360.
The highest common factor (HCF) of x and y is 60.

Find the values of n and m .

$$360 = 2^3 \times 3^2 \times 5$$

$$60 = 2^2 \times 3 \times 5$$

$$n = 1$$

$$m = 1$$

Answer $n = \dots\dots\dots 1$
 $m = \dots\dots\dots 1$ [2]

- (b) The square root of an integer z is equal to the cube of $3^2 \times 5^4$.

Write z as a product of its prime factors.

*

$$\sqrt{a} = a^{\frac{1}{2}}$$

$$\sqrt[3]{a} = a^{\frac{1}{3}}$$

$$\sqrt[m]{a} = a^{\frac{1}{m}}$$

$$\sqrt{z} = (3^2 \times 5^4)^3$$

$$\sqrt{z} = 3^6 \times 5^{12}$$

$$z = (3^6 \times 5^{12})^2$$

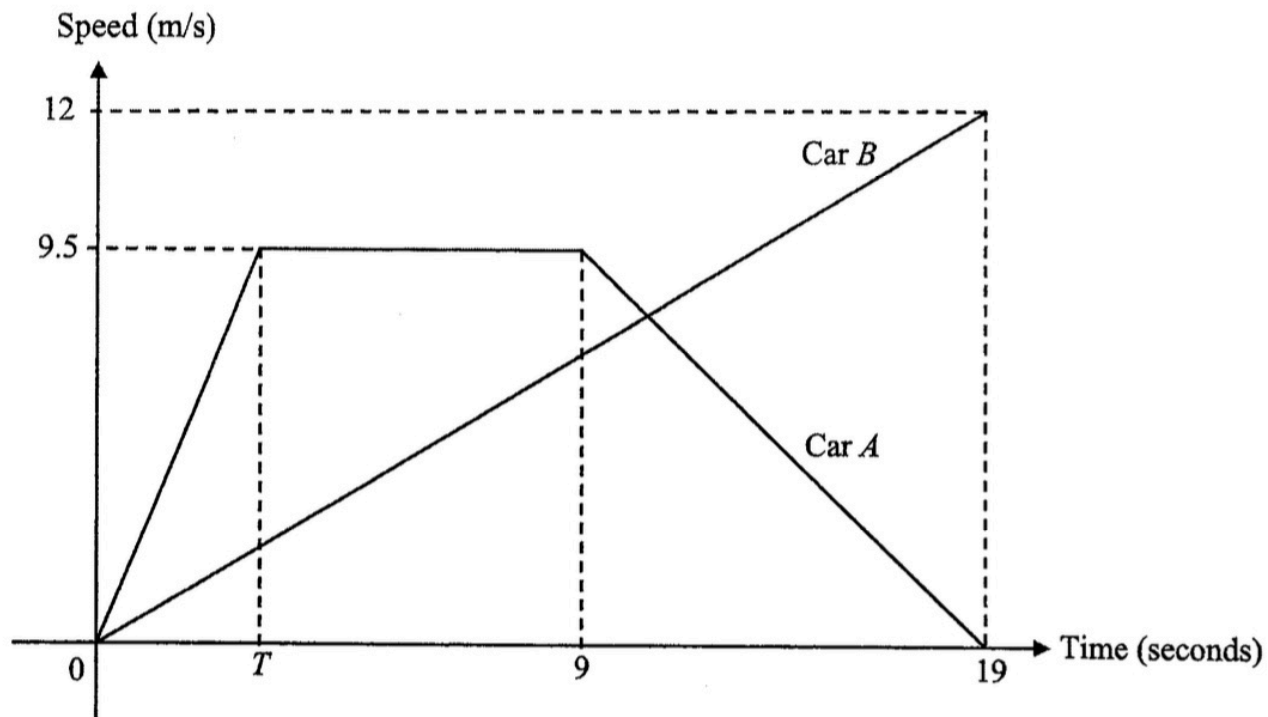
$$= 3^{12} \times 5^{24}$$

Answer $3^{12} \times 5^{24}$ [2]

25 Car A and B participate in a race.

Car A accelerates uniformly in the first T seconds and continue at a constant speed before decelerating to a rest in 10 seconds.

Car B accelerates uniformly to a speed of 12 m/s in 19 seconds.



(a) Find the deceleration of car A in the last 10 seconds.

Answer m/s^2 [2]

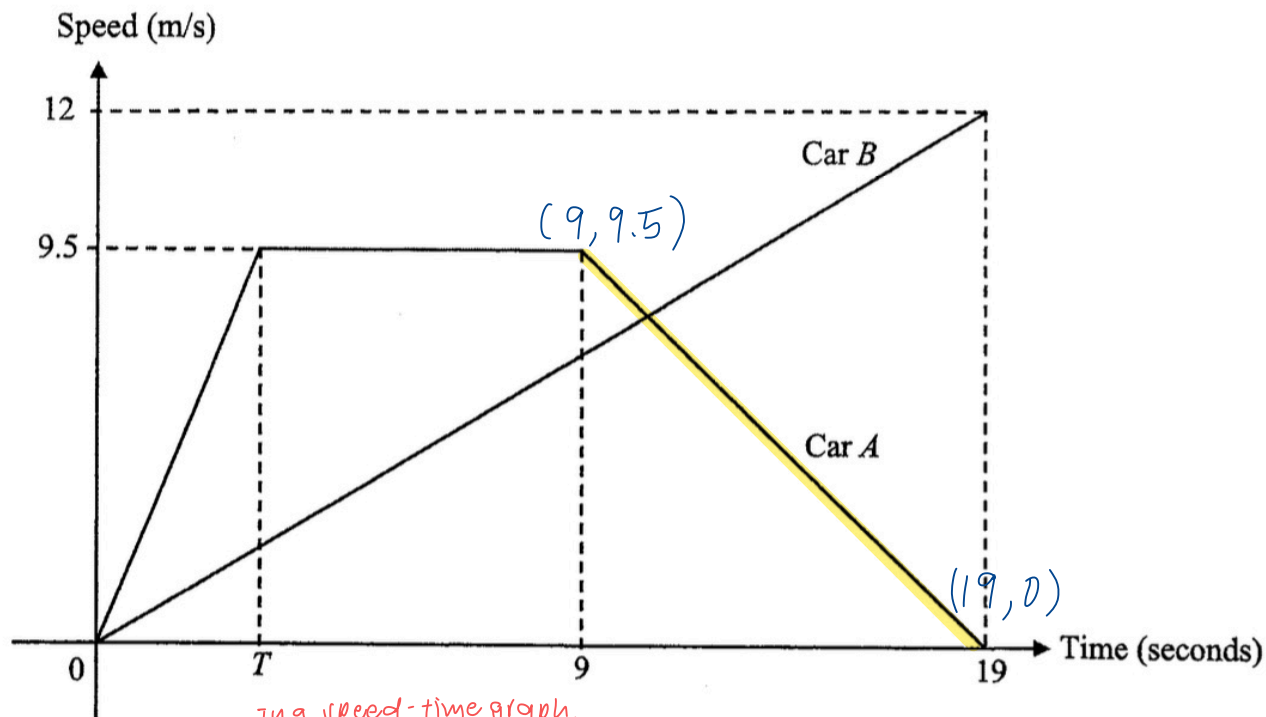
(b) Given that Car A and B complete the same total distance in 19 seconds.
Find the value of T .

Answer $T = \dots\dots\dots$ [3]

25 Car A and B participate in a race.

Car A accelerates uniformly in the first T seconds and continues at a constant speed before decelerating to a rest in 10 seconds.

Car B accelerates uniformly to a speed of 12 m/s in 19 seconds.



In a speed-time graph,
 $\rightarrow$ the gradient is the acceleration/deceleration.

(a) Find the deceleration of car A in the last 10 seconds.

$$\begin{aligned} \text{gradient} &= \frac{9.5 - 0}{9 - 19} \\ &= -0.95 \end{aligned}$$

Note:

since deceleration is describing decrease in speed, no need to include '-' in final answer

Answer 0.95 m/s² [2]

(b) Given that Car A and B complete the same total distance in 19 seconds. Find the value of T .

$\rightarrow$ distance = area under the graph

$$\begin{aligned} \text{Distance travelled by Car A} &= \frac{1}{2}[(9-T) + 19](9.5) \\ &= \frac{1}{2}(28-T)(9.5) \end{aligned}$$

$\rightarrow$ Area of trapezium
 $\frac{1}{2}(a+b)h$

$$\begin{aligned} \text{Distance travelled by Car B} &= \frac{1}{2} \times 19 \times 12 \\ &= 114 \text{ m} \end{aligned}$$

$$\begin{aligned} \frac{1}{2}(28-T)(9.5) &= 114 \\ 28-T &= 24 \\ T &= 4 \end{aligned}$$

Answer $T = \dots\dots\dots 4 \dots\dots\dots$ [3]